

Targeted SAT Math — Set 01

Student: Karina

Print this PDF. Work on paper. Calculator / Desmos is fine.

Do not print the last page for her (answer key).

This set is only the leaks from Test 5 and desk **B14**. It is not another full mixed packet. Items are instructor-written SAT-style questions, not official College Board stems.

SPR = student-produced response. Write a number (integer, decimal, or fraction).

1. Notes for this set only

Successive percent

- Each change **multiplies**. Do not add the percents.
- Up $p\%$ means $\times \left(1 + \frac{p}{100}\right)$. Down $p\%$ means $\times \left(1 - \frac{p}{100}\right)$.
- Up 10% then up 10% is $\times 1.1 \times 1.1 = 1.21$, a **21%** increase, not +20%.
- Up 10% then down 10% is $\times 1.1 \times 0.9 = 0.99$. You do **not** return to the start.

“Increase by $p\%$ ”

- Increase x by $p\%$ means $x \left(1 + \frac{p}{100}\right)$.
- Increase by 100% = double ($\times 2$). Increase by 400% = $\times 5$.
- “400% **of** x ” is a different sentence ($\times 4$).

Exponential period

- $A(1 + r)^{kt}$ completes k growth cycles in one unit of t .
- If t is in years, months per cycle = $12/k$.
- Example: exponent $2t \Rightarrow 2$ cycles per year $\Rightarrow$ every 6 months.

Tangent to a circle

- The radius to the contact point P is **perpendicular** to the tangent at P .

- Perpendicular slopes: $m_1 m_2 = -1$ (negative reciprocal). Flip the fraction and change the sign.
 - Vertical radius $\Rightarrow$ horizontal tangent (slope 0).
 - Write the tangent with point-slope **at** P , not at the center. Then test points in that line.
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2. Questions

1. (SPR) A club has 400 members. Membership increases 20% in year 1 and increases 20% again in year 2. How many members are there after 2 years?

Answer: ____

2. A \$50 price increases by 20%, then the new price decreases by 20%. What is the final price?

- A) \$50
- B) \$48
- C) \$52
- D) \$40

3. The result of increasing the quantity x by 300% is 80. What is the value of x ?

- A) $\frac{80}{3}$
- B) 20
- C) 240
- D) 320

4. (SPR) A population starts at 80 and grows by 25% each year. What is the population after 2 years?

Answer: ____

5. The function Q is defined by $Q(t) = 500(1.06)^{2t}$, where t is the number of years after 2020.

According to the model, Q increases by 6% every n months. What is the value of n ?

- A) 2
- B) 6
- C) 12
- D) 24

6. The function P is defined by $P(t) = 300(1.08)^{3t}$, where t is years after 2010. The model increases the population by 8% every n months. What is n ?

- A) 3
- B) 4

- C) 8
- D) 36

7. (SPR) A circle has center $(0, 0)$. A line is tangent to the circle at $(3, 4)$. What is the slope of the tangent line?

Answer: __

8. A circle in the xy -plane has center $(2, 3)$. Line ℓ is tangent to the circle at $(6, 6)$. Which point also lies on ℓ ?

- A) $(2, 3)$
- B) $(3, 10)$
- C) $(8, 9)$
- D) $(5, 3)$

9. (SPR) A circle has center $(-2, 1)$. A tangent line touches the circle at $(1, 5)$. The equation of the tangent line can be written as $y = mx + b$. What is the value of b ?

Answer: __

10. A radius of a circle runs from $(1, 2)$ to the point of tangency $(5, 4)$. What is the slope of the tangent line at $(5, 4)$?

- A) $\frac{1}{2}$
- B) 2
- C) -2
- D) $-\frac{1}{2}$

11. A circle has center $(4, 1)$. A line is tangent to the circle at $(4, 7)$. What is the slope of the tangent line?

- A) 0
- B) 1
- C) undefined
- D) 6

12. A jacket costs \\$40. The store marks it up 10%, then takes 10% off the **new** price. How much less than \\$40 is the final price?

- A) \\$0
- B) \\$0.40
- C) \\$4
- D) \\$8

3. Answer key (instructor)

Skip this page when you print the set for Karina.

#	Answer	Why
1	576	$400 \times 1.2 \times 1.2 = 576$. Adding 20% twice is the trap (560).
2	B, \$48	$50 \times 1.2 \times 0.8 = 48$. Same percent up and down does not cancel.
3	B, 20	Increase by 300% means $4x = 80$.
4	125	$80 \times (1.25)^2 = 125$. Same move as desk B14.
5	B, 6	Exponent $2t = 2$ cycles per year = every 6 months.
6	B, 4	Exponent $3t = 3$ cycles per year = every 4 months.
7	$-\frac{3}{4}$	Radius slope $\frac{4}{3}$; tangent is the negative reciprocal.
8	B, (3, 10)	Radius slope $\frac{3}{4}$, tangent slope $-\frac{4}{3}$. Line: $y - 6 = -\frac{4}{3}(x - 6)$. Choice A is the center.
9	$\frac{23}{4}$ or 5.75	Radius slope $\frac{4}{3}$, tangent slope $-\frac{3}{4}$. $y - 5 = -\frac{3}{4}(x - 1)$.
10	C, -2	Radius slope $\frac{1}{2}$, so $m_t = -2$. Product is -1 .
11	A, 0	Radius is vertical, so the tangent is horizontal.
12	B, \$0.40	$40 \times 1.1 \times 0.9 = 39.60$. Short of \$40 by 0.40.